

Technical Documentation

Process name:	Vehicle Contract Automation	Platform:	n8n
Type:	Learning Project / PoC	Domain:	AI-based data extraction
Author:	María José Jiménez	Status:	Not implemented in production

Problem	The billing team at a vehicle dealership opened each PDF contract and manually transcribed the vehicle data into a spreadsheet — a slow process prone to typing errors.																																																																						
Description	Workflow that reads vehicle purchase contracts in PDF format, extracts 8 fields using an AI model, and automatically logs them into a spreadsheet.																																																																						
Purpose	Eliminate manual transcription of contract data, reducing time and typing errors.																																																																						
Architecture	Google Drive source folder (PDFs) → n8n orchestration, 10 nodes → Google Gemini Information Extractor → Google Sheets append-or-update → Gmail final notification																																																																						
Workflow (node by node)	<table><tr><th>#</th><th>NODE</th><th>TYPE</th><th>WHAT IT DOES</th><th>RECEIVES → RETURNS</th><th>WHY IT IS NEEDED</th></tr><tr><td>1</td><td>When clicking "Execute workflow"</td><td>Manual start</td><td>This is the button that starts the process.</td><td>— → the process starts running</td><td>In this test it is started by hand; in real use it would trigger on its own, for example every day at a set time or as soon as a new contract arrives in the folder.</td></tr><tr><td>2</td><td>Search files and folders</td><td>Google Drive</td><td>Searches for and lists all the PDFs stored in the contracts folder.</td><td>Folder → 15 files found</td><td>This tells the process which contracts it needs to process in that run.</td></tr><tr><td>3</td><td>Loop Over Items</td><td>Loop</td><td>Takes the 15 contracts and processes them one by one, repeating the following steps for each one.</td><td>15 files → 1 contract at a time</td><td>If one contract fails, it doesn't affect the others. Also, once all of them have been processed, an email is sent notifying that the process has finished.</td></tr><tr><td>4</td><td>Download file</td><td>Google Drive</td><td>Downloads the PDF contract that is being processed at that moment.</td><td>File name → downloaded PDF file</td><td>To be able to read the contract's content, it must first be downloaded.</td></tr><tr><td>5</td><td>Extract from File</td><td>Extract From PDF</td><td>Converts the PDF content into readable text.</td><td>PDF file → contract text</td><td>The AI model that extracts the data needs text; it cannot read a PDF directly.</td></tr><tr><td>6</td><td>Code in JavaScript</td><td>Text cleanup</td><td>Cleans and organizes the contract text before sending it to the AI.</td><td>Unorganized text → clean text</td><td>Improves the accuracy with which the AI identifies the vehicle data.</td></tr><tr><td>7</td><td>Wait</td><td>Wait</td><td>Adds a short pause before querying the AI with the next contract.</td><td>— → a brief wait</td><td>The AI only allows a limited number of requests per minute; this pause prevents the process from failing by exceeding that limit.</td></tr><tr><td>8</td><td>Information Extractor</td><td>AI (Gemini)</td><td>Reads the contract text and extracts the 8 vehicle data fields.</td><td>Contract text → the 8 vehicle data fields, already organized</td><td>The AI interprets the text extracted from the document to extract the vehicle data.</td></tr><tr><td>9</td><td>Append or update row in sheet</td><td>Google Sheets</td><td>Saves the contract data as a new row in the spreadsheet; if that vehicle was already registered, it updates its row instead of creating a duplicate.</td><td>The 8 data fields → a row in the sheet</td><td>Prevents duplicate contracts if the process is run more than once.</td></tr><tr><td>10</td><td>Send a message</td><td>Gmail</td><td>Sends an email notifying that all contracts have finished being processed.</td><td>End of process → email sent</td><td>This way the billing team doesn't have to check the sheet to know whether the process has finished.</td></tr></table>					#	NODE	TYPE	WHAT IT DOES	RECEIVES → RETURNS	WHY IT IS NEEDED	1	When clicking "Execute workflow"	Manual start	This is the button that starts the process.	— → the process starts running	In this test it is started by hand; in real use it would trigger on its own, for example every day at a set time or as soon as a new contract arrives in the folder.	2	Search files and folders	Google Drive	Searches for and lists all the PDFs stored in the contracts folder.	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Technical decision: AI model	Role: Gemini is the model used by the Information Extractor node (step 8). Based on the contract text, it identifies and returns the 8 required fields in structured format, replacing manual transcription. Selection criteria: • Cost: the free tier is sufficient for the workload, with no need to purchase a paid plan. • Context window: allows multi-page contracts to be processed in a single call, with no need to split them.
Results and potential impact	The automation eliminates manual transcription of contract data, reducing the risk of typing errors and freeing up time for the billing team to focus on higher-value tasks.